Reproducing MCTS-based Information Seeking

Yubo Wang · INFO 7375 · Assignment 2&3

1.About the Papers

UoT (Hu et al., NeurIPS 2024)

- Tree search + information gain → pick best question
- DX dataset (104 patient samples): 80–83% SR (Llama 70B / Mixtral / GPT-4o)

MISQ-HF (Chopra & Shah, NeurIPS 2025)

- MCTS + hierarchical feedback from past successes
- Up to 98% SR (Llama 70B), 99% (GPT-4o), ~10× fewer LLM calls

Our Reproduction

```
Paper models:
  Llama 3.3 70B
  Mixtral 8×7B (~47B)
  GPT-4o
```

→

```
My models:
  llama3.1:8b
  qwen2.5:7b
  (~10× smaller)
```

2.Key Ideas

UoT: Information Gain

```
R_IG = H(p_a, p_n) / (1 + |p_a-p_n| / λ)
```

- Score by how evenly question splits diseases

  (50-50 split = highest score, 100-0 = lowest)
- λ = 0.4 penalizes unbalanced splits

```
R_acc = R_IG + parent's R_acc

R_exp = p_a·R(YES) + p_n·R(NO)
```

→ pick question with highest R_exp
(weighted avg: consider both YES and NO branches)

MISQ-HF: UCT + Feedback

```
UCT = R̄(v) + C√(ln N_p / N_v) + B_k(v)
```

- C = 0.2 balances explore vs exploit

  (R̄ = avg reward, √ = try less-visited, B_k = feedback)
- Shared tree: 1 tree for all 104 patients

```
B_k(v) += β · R_total · γ^depth
```

- On success → bonus to path nodes

  (HF = learn from past: good paths get reused)
- γ = 0.9: early questions get more bonus
- Cluster by embedding (cos_sim ≥ 0.9)

3.Data & DP

DX Dataset

- 104 samples, 5 diseases, max 6 turns
- Self-play: same model = doctor + patient
- Closed set (Ω-aware), temp = 0

5 Diseases in DX:

HFMD · Infantile diarrhea
Allergic rhinitis · URTI · Pneumonia

DP Results: Why Tree Search Is Needed

	SR (%)	MSC
Ours:		
llama 8B	74.04	4.99
qwen 7B	16.35	4.12
Paper:		
Llama 70B	88.46	3.15
Mixtral 47B	50.00	3.50
GPT-4o	73.07	3.48

Our llama (74%) between Mixtral (50%) and Llama 70B (88%) → setup works
qwen (16%) → needs tree search badly

4.Implementing UoT & MISQ

UoT: QNode ↔ RNode Tree

```
RNode(ROOT, Ω=[d1..d5])
├── QNode("Fever?")
│   ├── RNode(YES) → [d1,d2] terminal
│   └── RNode(NO)  → [d3,d4,d5]
│       ├── QNode("Cough?")
│       └── QNode("Breathing?")
└── QNode("Stomach pain?")
```

- Depth 3, m=3 candidates per node
- 60% turns → tree search, 40% → guessing
- Fresh tree per patient

MISQ: 4-Step MCTS

Selection	UCT picks best child
Expansion	LLM generates new Qs for node
Simulation	Random rollout to estimate value
Backprop	Update rewards to root

Key Difference

UoT → fresh tree per patient
MISQ → ONE shared tree for all 104

5.Pitfalls During Reproduction

#1 Parsing = 40% of Work

- LLM output format is unreliable
- Names misspelled, labels missing
- qwen: Chinese punctuation
- Solution: two-pass parser
+ fuzzy matching
+ auto-fill + 3 retries

#2 |Ω|=5 Too Small

250×

QGC reduction (paper: ~10×)

- Tree fully expanded after 1 patient
- 103 patients → 0 new LLM calls
- Paper's 10× on FloDial (153 faults)
- 5 diseases = trivially small for MCTS

#3 Feedback Adds Noise

62.5%

MISQ-HF (vs MISQ 80.8%)

- Tree complete → can only re-rank
- τ=0.9 → each patient = own cluster
- Bonus never shared between patients
- Paper Mixtral: same pattern (77≈63%)

6.Results

Ours (7–8B)

Method	Model	SR%	MSC	QGC
DP	llama 8B	74.04	4.99	-
DP	qwen 7B	16.35	4.12	-
UoT	llama 8B	80.77	3.00	8.0
UoT	qwen 7B	80.77	3.24	4.0
MISQ	llama 8B	80.77	3.00	0.03
MISQ-HF	llama 8B	62.50	2.71	0.05

Paper (DX, Ω-aware)

Method	Model	SR%	MSC	QGC
DP	Llama 70B	88.46	3.15	-
UoT	Llama 70B	79.80	1.65	0.77
MISQ	Llama 70B	92.30	1.28	0.48
MISQ-HF	Llama 70B	98.07	1.84	0.04
DP	Mixtral 47B	50.00	3.50	-
UoT	Mixtral 47B	76.92	1.43	0.45
MISQ-HF	GPT-4o	99.03	2.19	0.03

UoT: Matches Paper ✓

Our 8B: 80.77%
Llama 70B: 79.80%
Mixtral: 76.92%

MISQ = UoT (tree too small)

Paper: 92% > 80% (MISQ > UoT)
Ours: 80% = 80%
→ |Ω|=5 saturates tree

MISQ-HF: Biggest Gap

Paper: 98% (70B)
Ours: 62% (8B)
Mixtral: 77% ≈ 63%
→ weak model + small Ω

7.Takeaways

Build Incrementally

DP → UoT → MISQ → MISQ-HF
Verify each step first

40% Time on Parsing

Small models ≠ GPT-4 output
Robust parser is essential

Use $|\Omega| \geq 15$

5 diseases too small for MCTS
Try MedDG (15) or larger

Lessons

Directional findings: tree search helps, shared trees save compute.
Feedback needs strong models + large search space — papers test under ideal conditions.